# Coupon Collector's Problem

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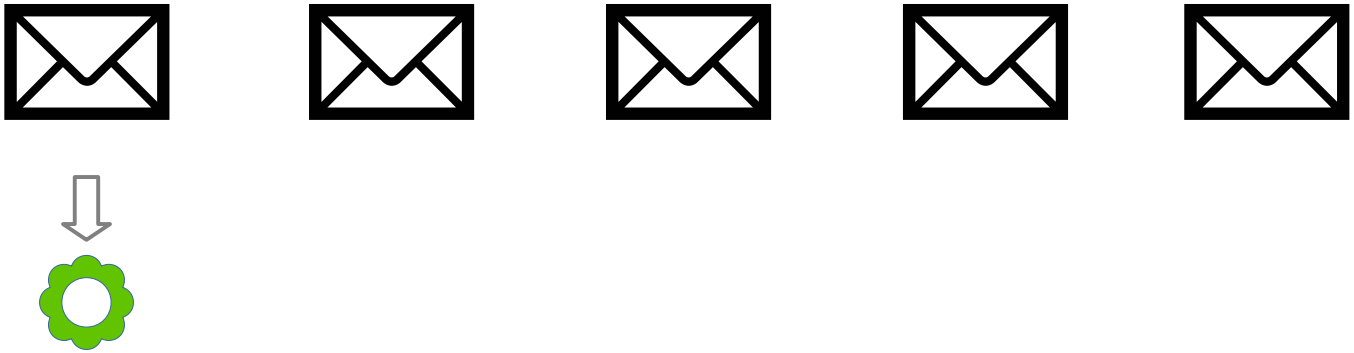
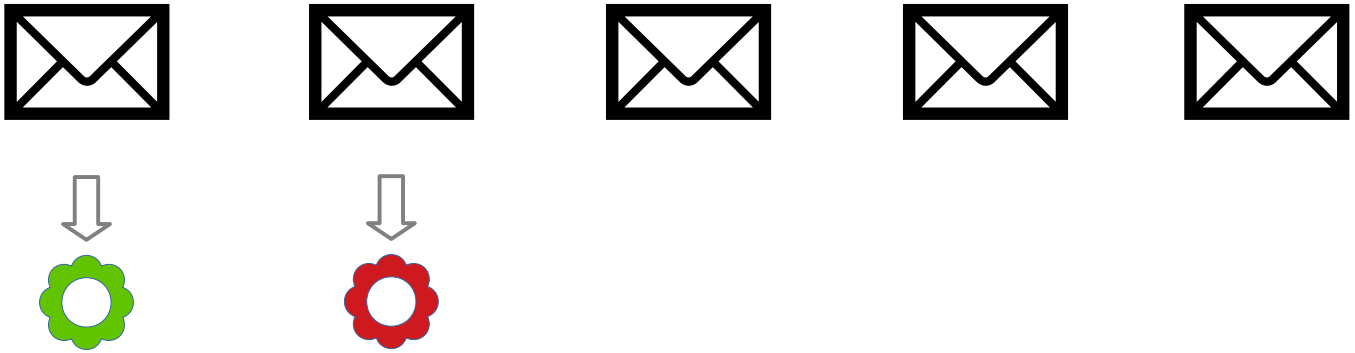
# Our memories (2005) – Hello Kitty Magnets

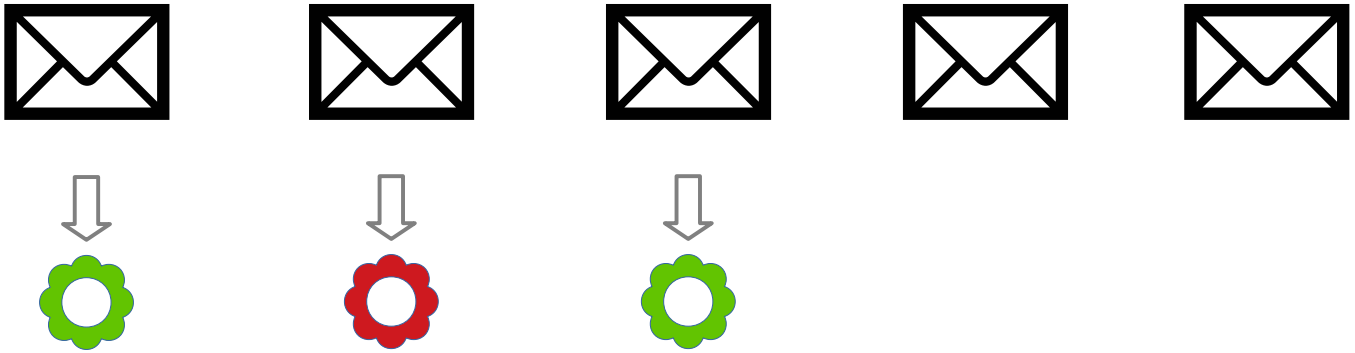
https://a.rimg.com.tw/s1/7/f7/89/10061029175177_650.jpg

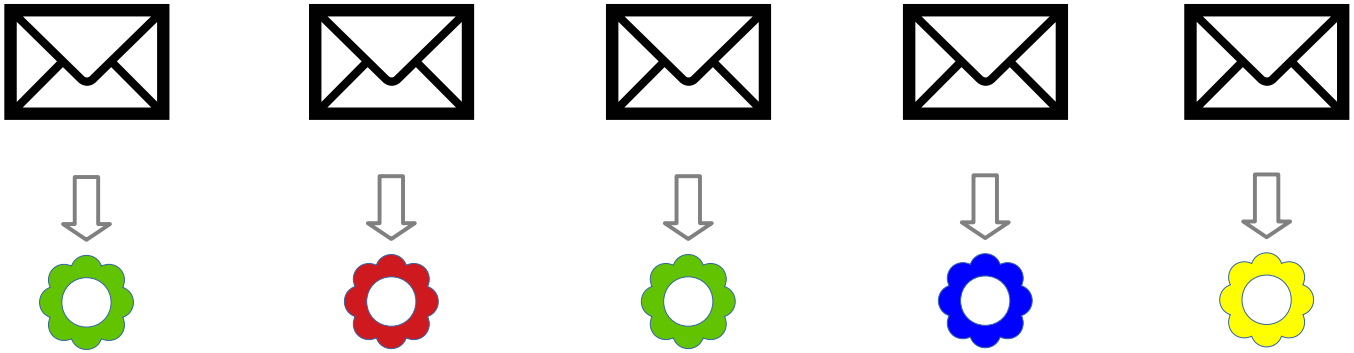
# Outline


1. Coupon Collector's Problem
2. Idea from Geometric Distributions
3. Conditional Expectations









# Coupon Collector's Problem

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- Have you thought about how much you should roughly pay for them?

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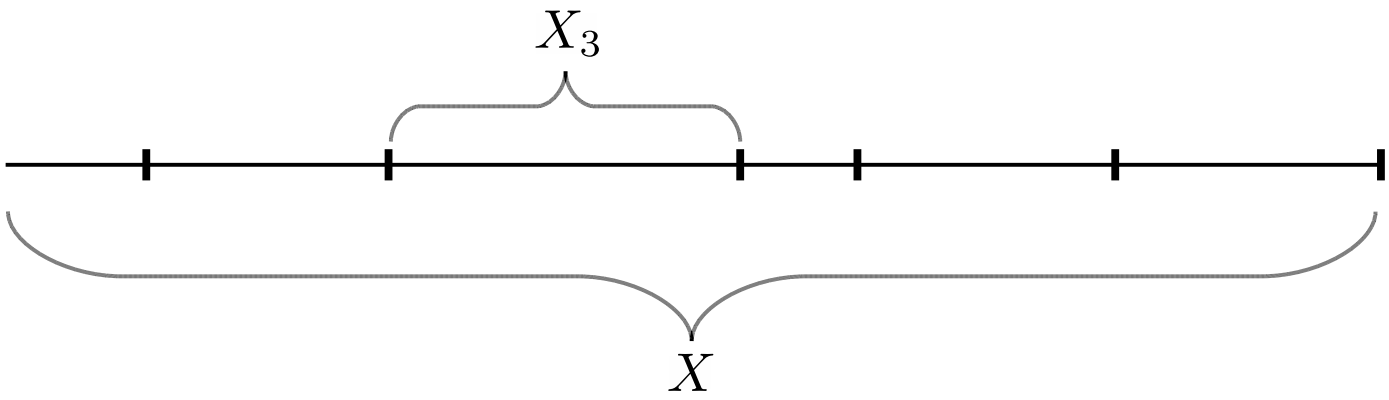
- Have you thought about how much you should roughly pay for them?
  - You need to pay NT$ 77 to get one coupon.

# Coupon Collector's Problem (stage by stage)



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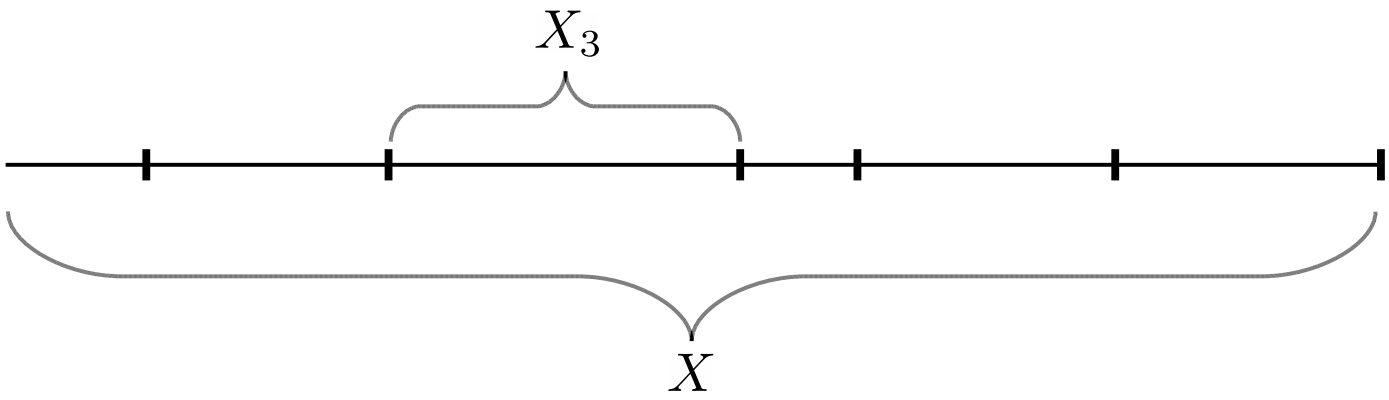


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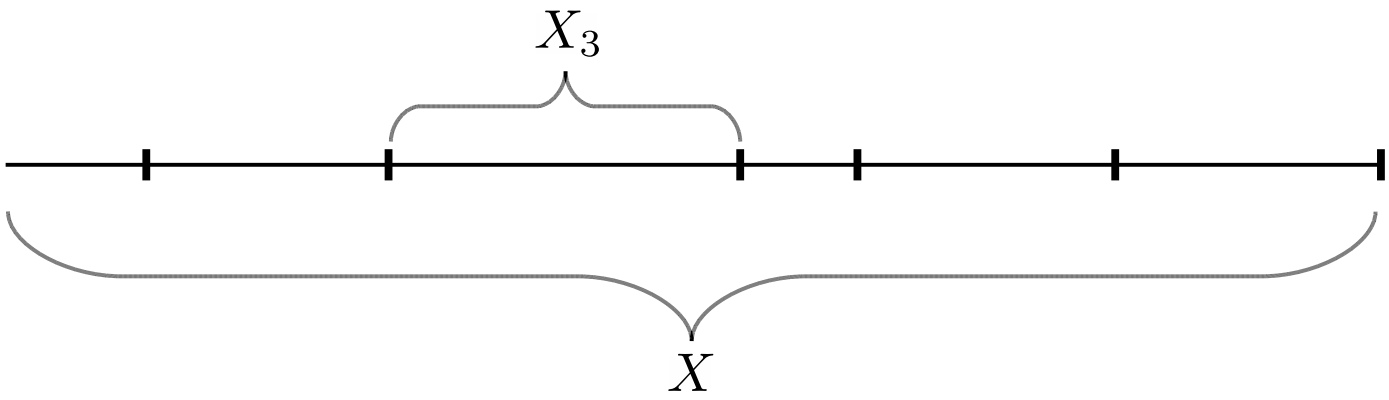


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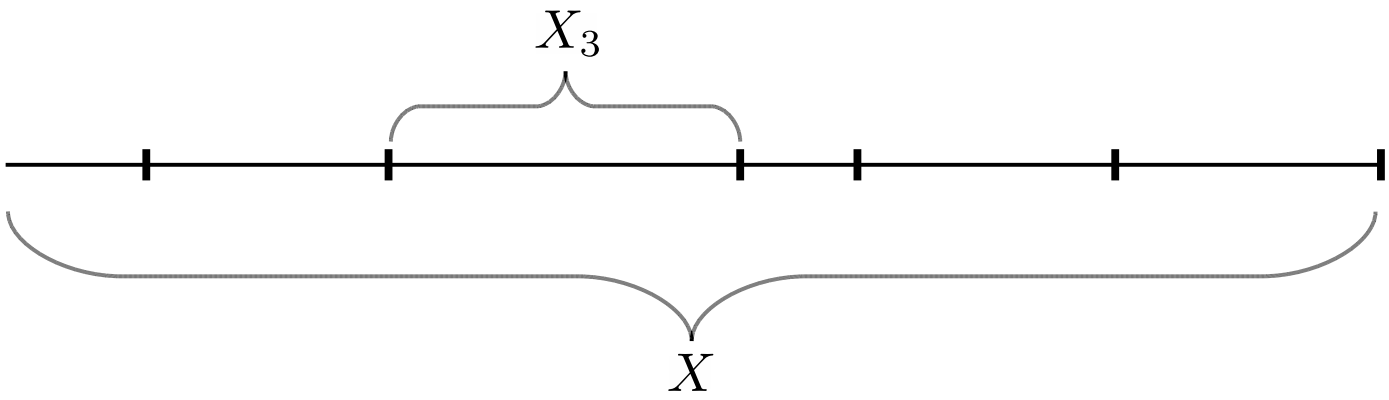


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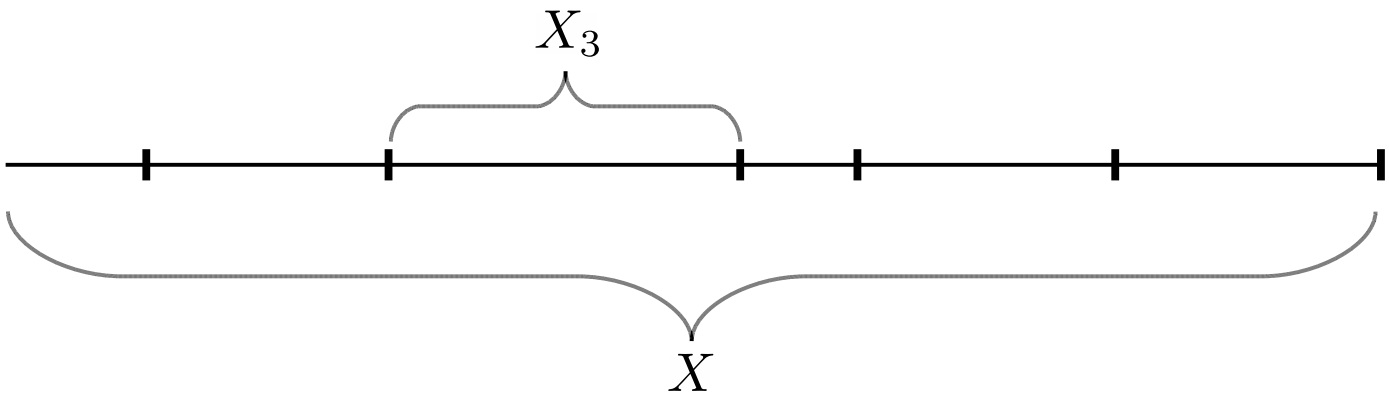


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  - $X_i$: Geometric random variables

# Geometric Distribution

**Consider the scenario:**

- flip a coin until it lands on a head

- What's the distribution of the number of flips?

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**Geometric random variable**

For a geometric random variable $X$ with parameter $p$, we have

$$\Pr[X = n] = (1-p)^{n-1}p.$$

for $n = 1, 2, \ldots$

# Properties of a geometric random variable

## Memorylessness

Let $X$ be a geometric random variable $X$ with parameter $p > 0$, we have

$$\Pr[X = n + k \mid X > k] = \Pr[X = n]$$

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## Expected Value

Let $X$ be a geometric random variable $X$ with parameter $p > 0$, we have

$$\mathbb{E}[X] = \frac{1}{p}.$$

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&= (1-p)^{n-1}p \\
&= \Pr[X = n].
\end{aligned}
$$

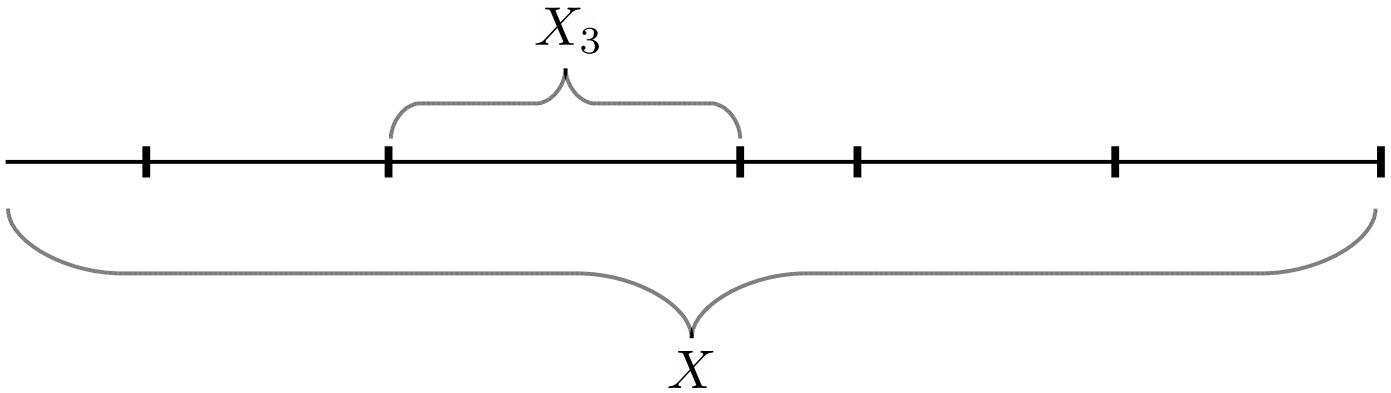
# The mean of a geometric r.v. $X(p)$

$$\begin{aligned}\mathbb{E}[X] &= \sum_{j=1}^{\infty} j \Pr[X=j] \\ &= \sum_{j=1}^{\infty}\sum_{i=1}^{j} \Pr[X=j] \\ &= \sum_{i=1}^{\infty}\sum_{j\geq i} \Pr[X=j] \\ &= \sum_{i=1}^{\infty} \Pr[X\geq i].\end{aligned}$$

$$\Pr[X\geq i] = \sum_{k=i}^{\infty}(1-p)^{k-1}p = (1-p)^{i-1}.$$

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=1}^{\infty}(1-p)^{i-1} \\ &= \frac{1}{1-(1-p)} \\ &= \frac{1}{p}.\end{aligned}$$

# Back to the Coupon Collector's Problem

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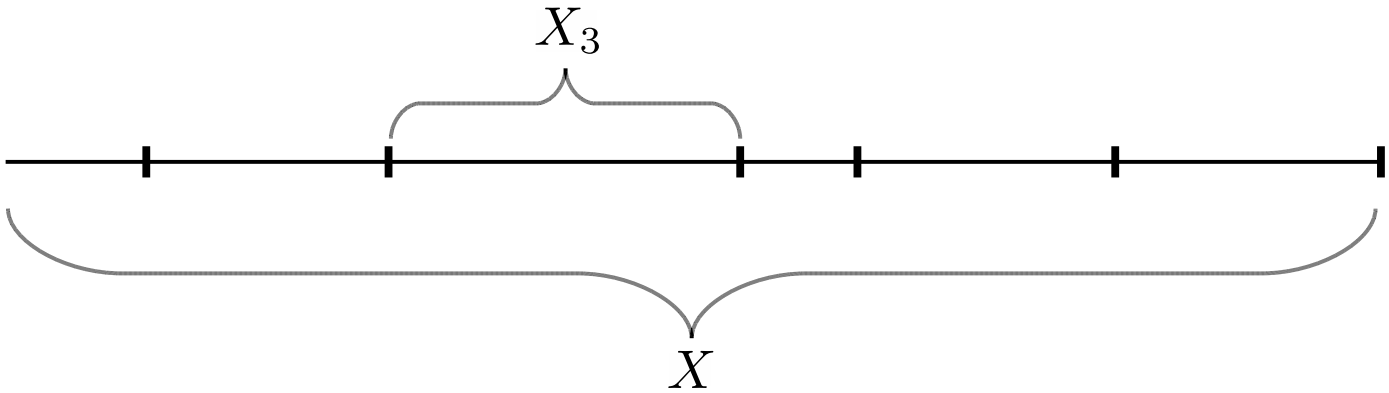


- When exactly $i-1$ coupons have been collected, the probability of obtaining a new one is

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- Since each $X_i$ is a geometric random variable, we gave

$$\mathbf{E}[X_i] = \frac{1}{p_i} = \frac{n}{n-i+1}.$$

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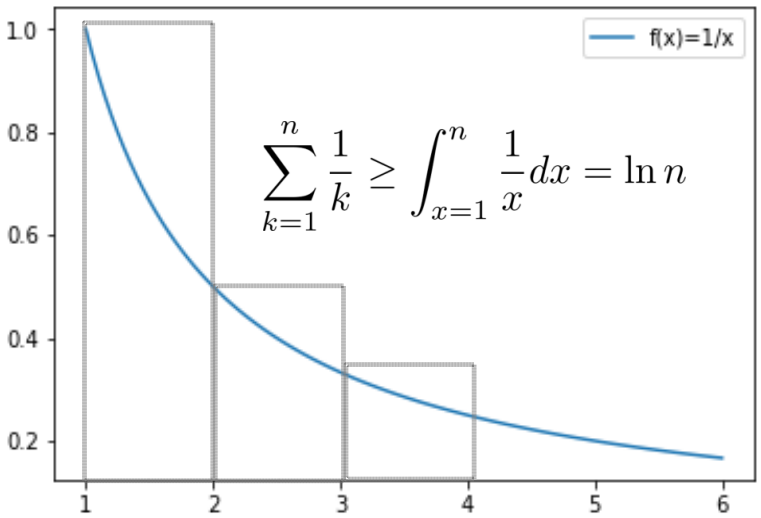


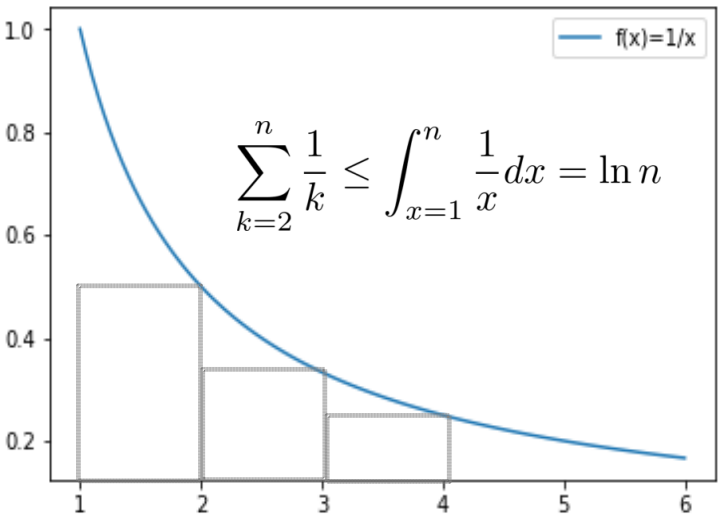

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- How likely is this to happen?
  - $\Pr[X \geq 2R] \leq \frac{1}{2}$ (Markov Inequality).

- Let $X$ be a random variable that assumes only non-negative values. Then, for all $a > 0$,

$$\Pr[X \geq a] \leq \frac{\mathbf{E}[X]}{a}.$$

**Markov Inequality**

Andrei Andreyevich Markov (Wikipedia)
1856–1922

# This can be very likely to happen actually...

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- How likely is this to happen?
  - $\Pr[X \geq 2R] \leq \frac{\pi^2}{6 \ln^2 34} < 0.1323$ (Chebyshev's Inequality).

<u>Theorem</u> [Chebyshev's Inequality]. For any $a > 0$,

$$\Pr[|X - \mathbf{E}[X]| \geq a] \leq \frac{\mathbf{Var}[X]}{a^2}.$$

Learn the coupon collector's problem well!
Pay me NT$ 30,000 and you get them all...

# Variance of a geometric random variable

## Variance

Let $X$ be a geometric random variable $X$ with parameter $p > 0$, we have

$$\mathbf{Var}[X] = \frac{1-p}{p^2}.$$

Let $X_1, X_2, \ldots, X_n$ be $n$ independent geometric random variable $X$ with parameters $p_1, p_2, \ldots, p_n > 0$ respectively. Let $X = \sum_{i=1}^{n} X_i$, then we have

$$\mathbf{Var}[X] = \sum_{i=1}^{n} \frac{1-p_i}{p_i^2} < \sum_{i=1}^{n} \frac{1}{p_i^2}.$$

$$\begin{aligned}\mathbf{Var}[X] = \sum_{i=1}^{n} \mathbf{Var}[X_i] < \sum_{i=1}^{n} \left(\frac{1}{p^2}\right) &= \sum_{i=1}^{n} \left(\frac{n}{n-i+1}\right)^2 \\ &= n^2 \sum_{i=1}^{n} \left(\frac{1}{i}\right)^2\end{aligned}$$

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&= n^2 \sum_{i=1}^{n} \left(\frac{1}{i}\right)^2 \\
&\leq \frac{\pi^2 n^2}{6}. \quad \text{(Leonhard Euler (1740))}
\end{aligned}$$

**Exercise**

What is $\mathbb{E}[X]$ when you have already collected 10 types of coupons?

# Outline

1 Coupon Collector's Problem

2 Idea from Geometric Distributions

3 Conditional Expectations

# Conditional Expectation

## Definition

$$\mathbb{E}[Y \mid Z = z] = \sum_{y} y \Pr[Y = y \mid Z = z].$$

- Example of two dice.
  - $X_1$: the number showing on the first die
  - $X_2$: the number showing on the second die
  - $X = X_1 + X_2$

$$\mathbb{E}[X \mid X_1 = 2] = \sum_{x=3}^{8} x \cdot \frac{1}{6} = \frac{11}{2}.$$

# Conditional Expectation (contd.)

**Lemma (Law of Total Expectation)**

For any random variables $X$ and $Y$,

$$\mathbb{E}[X] = \sum_y \Pr[Y = y]\, \mathbb{E}[X \mid Y = y].$$

$$\sum_y \Pr[Y = y] \cdot \mathbb{E}[X \mid Y = y] =$$

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&= \sum_x \sum_y x \Pr[X = x \mid Y = y] \cdot \Pr[Y = y] \\
&= \sum_x \sum_y x \Pr[\{X = x\} \cap \{Y = y\}]
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&= \sum_x \sum_y x \Pr[\{X = x\} \cap \{Y = y\}] \\
&= \sum_x x \Pr[X = x] = \mathbb{E}[X].
\end{aligned}$$

# Remark

By marginalization of the joint distribution,

$$\sum_{y} \Pr[\{X = x\} \cap \{Y = y\}] = \Pr[X = x].$$

# Conditional Expectation

**Lemma**

For any finite collection of discrete random variables $X_1, X_2, \ldots, X_n$ with finite expectations and for any random variable $Y$,

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i \,\middle|\, Y = y\right] = \sum_{i=1}^{n} \mathbb{E}[X_i \mid Y = y].$$

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- It may seem a bit strange to you to see:
  $\mathbb{E}[Y \mid Z]$: regarded as a random variable $f(Z)$.
  - It takes on the value $\mathbb{E}[Y \mid Z = z]$ when $Z = z$.
- In the previous example,

$$\mathbb{E}[X \mid X_1] = \sum_x x \cdot \Pr[X = x \mid X_1] = \sum_{x=X_1+1}^{X_1+6} x \cdot \frac{1}{6} = X_1 + \frac{7}{2}.$$

- So it makes sense that

$$\mathbb{E}\big[\mathbb{E}[X \mid X_1]\big] = \mathbb{E}\left[X_1 + \frac{7}{2}\right] = \frac{7}{2} + \frac{7}{2} = 7.$$

# Conditional Expectation (contd.)

**Theorem**

$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid Z]]$.

- *Proof.*

$$\mathbb{E}[\mathbb{E}[Y \mid Z]] = \sum_z \mathbb{E}[Y \mid Z = z] \cdot \Pr[Z = z] = \mathbb{E}[Y].$$

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- The last equality follows from the law of total expectation.

# Discussions